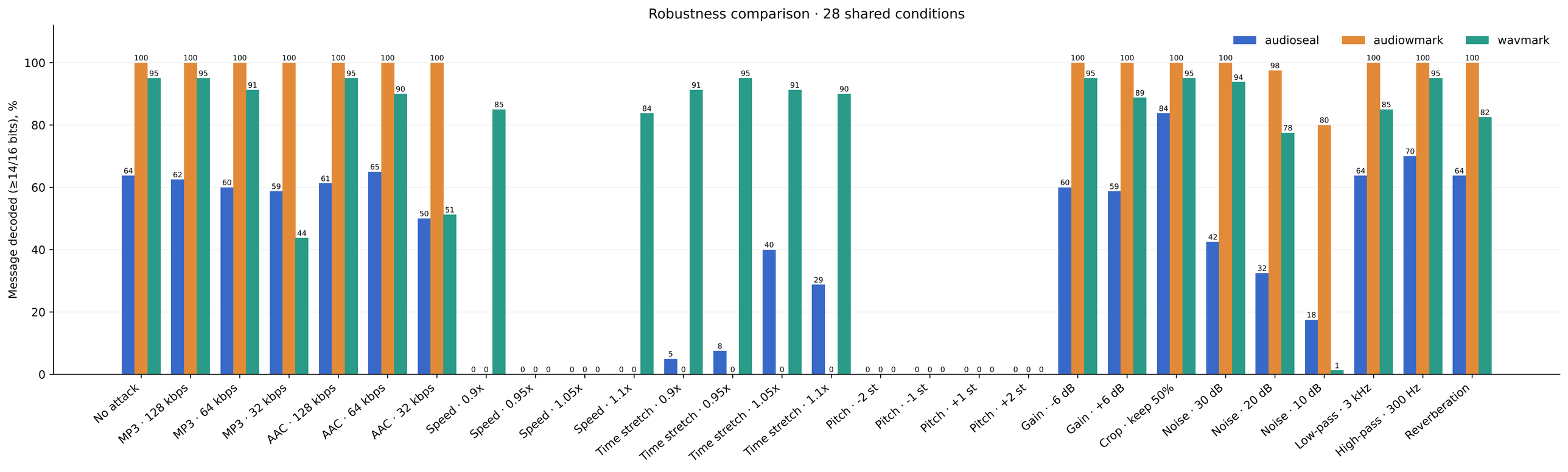
Robustness comparison · 28 shared conditions
audioseal
audiowmark
wavmark
Message decoded (≥14/16 bits), %
0
20
40
60
80
100
No attack
MP3 · 128 kbps
MP3 · 64 kbps
MP3 · 32 kbps
AAC · 128 kbps
AAC · 64 kbps
AAC · 32 kbps
Speed · 0.9x
Speed · 0.95x
Speed · 1.05x
Speed · 1.1x
Time stretch · 0.9x
Time stretch · 0.95x
Time stretch · 1.05x
Time stretch · 1.1x
Pitch · -2 st
Pitch · -1 st
Pitch · +1 st
Pitch · +2 st
Gain · -6 dB
Gain · +6 dB
Crop · keep 50%
Noise · 30 dB
Noise · 20 dB
Noise · 10 dB
Low-pass · 3 kHz
High-pass · 300 Hz
Reverberation